CISC 6080 · FORDHAM UNIVERSITY · CAPSTONE PROJECT

Affect-Diff

Multimodal Emotion Recognition via Causal-Diffusion Bridge

The problem in one sentence: Standard models achieve 66 % accuracy on CMU-MOSEI by **never predicting Fear, Disgust, or Surprise** three of six Ekman emotions at all.

What
6-class emotion recognition from text, audio & video (CMU-MOSEI)

Why hard
Happy = 65.9 % of data. Three classes under 2 % each.

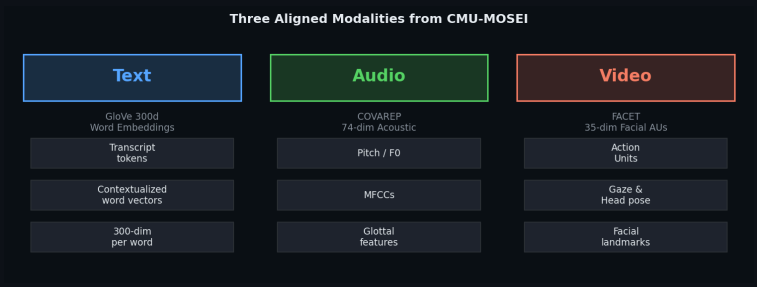
Our approach
Causal graph + β-VAE + stop-gradiented DDPM prior

CMU-MOSEI · NOTEARS Causal Graph · Diffusion Models · Class Imbalance · β-VAE

Ankit Sanjyal · asanjyal56@fordham.edu

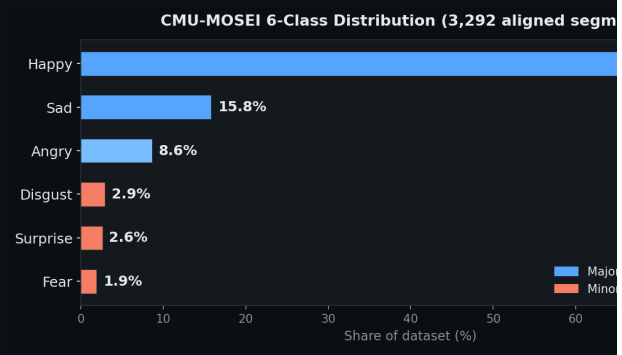
The Dataset: CMU-MOSEI

CMU-MOSEI contains **23,453 utterances** from 1,000 YouTube speakers, each aligned across three modalities. We use a **3,292-segment** subset with clean tri-modal alignment.



Each segment is labeled for **6 Ekman emotions** (Happy, Sad, Angry, Disgust, Surprise, Fear) — not sentiment polarity. Labels are independent per emotion; multi-label possible.

Why 3,292 of 23,453? — Strict **tri-modal alignment**: every segment needs valid GloVe + COVAREP + FACET arrays with matching temporal intervals. Misaligned or partially missing segments are discarded, leaving ~14 % of the corpus. This is a **data-limited regime by design**, not by choice — and our sentiment results show the model scales: 7× more data → **+90 % BalAcc**.



The core challenge: Happy alone is 65.9 % of labels. Fear, Disgust, a
collectively represent under 7 % yet all three are clinically meaningfu
that a real system must detect.

Why Existing Models Fail

TWO COMPOUNDING FAILURE MODES

① Majority-class collapse

Under severe imbalance, every fusion model maximises accuracy by predicting Happy / Sad for everything. A "predict Happy always" classifier scores **66 % accuracy** while being completely useless for emotion recognition.

② Modality collapse

Models over-rely on text and ignore audio / video the modalities that carry prosody and facial expression. When text degrades, performance collapses entirely.

> Focal loss, class-weighted loss, post-hoc causal correction (CausalMER), and diffusion imputation (IMDer) each address one axis none address **both simultaneously in the forward pass**.

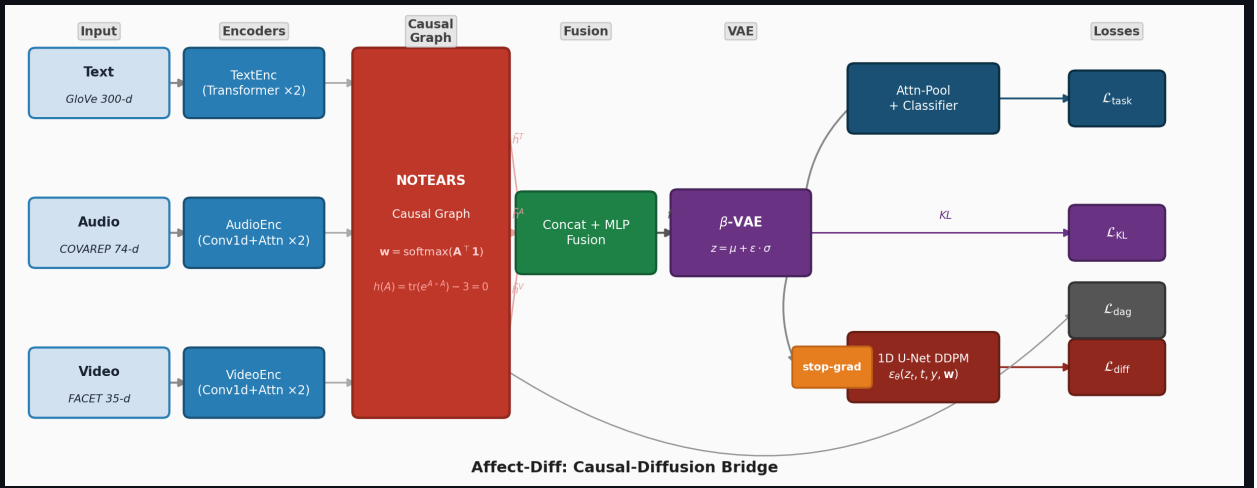
FIVE BASELINES SAME BLIND SPOT

Method	Year	Val-BalAcc ↑	Tes
TFN	2017	0.248	0.6
MulT	2019	0.278	0.6
MISA	2020	0.278	0.6
MMIM	2021	0.266	0.6
TETFN	2022	0.324	0.6
Affect-Diff		**0.384**	**0.6**

All five produce **zero F1** on Fear, Disgust, and Surprise re-implemented under conditions (same split, focal loss, label smoothing).

Val-BalAcc = macro-average recall across all 6 classes. **This is the metric tha gamed by majority-class predictions.**

Architecture Overview: Causal-Diffusion Bridge

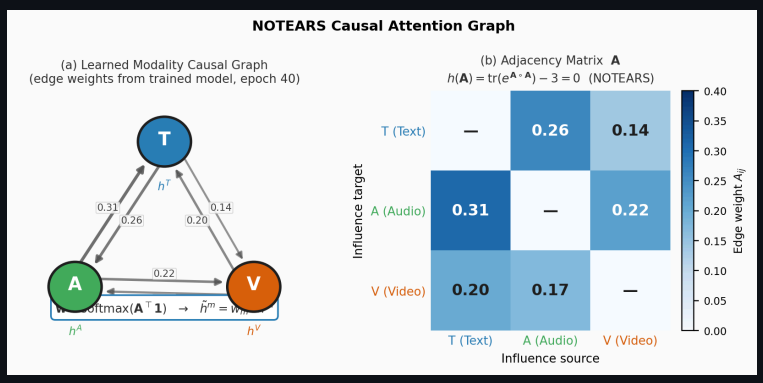


NOTEARS Causal Graph
Learns directed edges over {T, A, V}. Column sums → weights **w** that gate each modality before fusion. Same **w** conditions the U-Net.

β-VAE Bottleneck
Compresses fused representation to latent **z**. Free-bits KL prevents posterior collapse on rare classes.

Stop-Gradiented DDPM
1D U-Net shapes latent manifold from **sg(**
regularizes without conflicting with classi
gradients.

Causal Attention Graph: Structure and Learned Weights



WHAT IT LEARNS

A directed acyclic graph over 3 modality nodes {T, A, V} using scaled do
attention, constrained to be a DAG via the **NOTEARS penalty**:

```
h(A) = tr(e^(A∘A)) − 3 = 0
```

WHY IT MATTERS

Standard fusion treats all modalities equally. The causal graph learns wh
to trust per sample, gating $\tilde{h}^m = w_m \cdot h^m$ before fusion.

The same weights **w** condition the diffusion U-Net, making the denoiser
aware.

Removing NOTEARS (→ Gumbel-Softmax) drops val-BalAcc by **−0.059 (−**
confirming the acyclicity constraint adds real signal beyond simple sparsit

Affect-Diff: Three Jointly Trained Mechanisms

① NOTEARS CAUSAL GRAPH

Learns a DAG over {T, A, V}. Column sums produce importance weights that gate each modality's hidden sequence **before** fusion not post-hoc. Same weights condition the diffusion denoiser.

$$w = \mathrm{softmax}(A^T \cdot 1) \qquad \tilde{h}^m = h^m \cdot w_m \qquad h(A) = \mathrm{tr}(e^{(A \circ A)}) - 3 = 0$$

② B-VAE BOTTLENECK

Compresses the fused representation to a regularized latent $z \in \mathbb{R}^{128}$. Free-bits objective prevents posterior collapse on majority classes.

$$z = \mu + \varepsilon \cdot \sigma \qquad L_{KL} = \beta \cdot \max(0, KL - \lambda)$$

③ STOP-GRADIENTED DDPM PRIOR

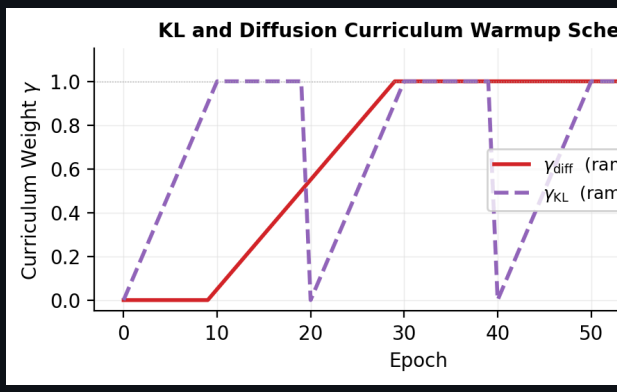
A 1D U-Net denoiser shapes the latent manifold. Gradient is **blocked** from reaching the encoder regularizes without conflicting with classification loss.

$$z_{diff} = sg(z) \qquad L_{diff} = \|\varepsilon - \varepsilon_\theta(z_t, t, y, w)\|^2$$

Combined loss:

$$L = L_{task} + \gamma_{kl} \cdot L_{KL} + \gamma_{diff} \cdot L_{diff} + \lambda_{dag} \cdot h$$

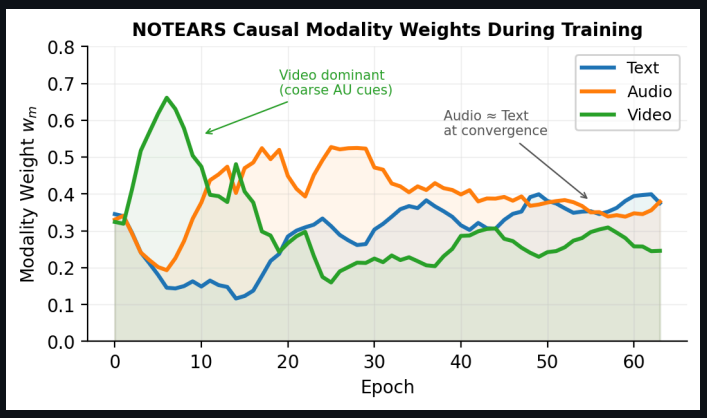
Each γ is ramped up via curriculum warmup classifier dominates first, au
phase in gradually.



Classifier-only (ep 0–9) → KL ramps in (ep 0–30) → diffusion ramps in (ep 9
Prevents early posterior collapse while letting the classifier stabilize.

What the Model Actually Learns

CAUSAL MODALITY DYNAMICS



Epoch 10: Video dominates (V≈0.58) facial AU cues guide coarse class boundaries early.
Epoch 40+: Audio ≈ Text (A≈0.38, T≈0.40), Video recedes fine-grained prosody and language take over.

This is **non-trivial**: the graph doesn't converge to uniform or text-dominant weights it discovers temporal structure in which modality is most informative.

ABLATION: WHAT EACH COMPONENT ADDS

Configuration	Val-BalAcc	Δ vs Full
Full Model (3 seeds)	**0.384 ± 0.000**	
No Stop-Gradient	0.291	−0.093
No Diffusion Prior	0.292	−0.092
No NOTEARS (Gumbel)	0.325	−0.059
No Causal Graph	0.334	−0.050
No VAE (deterministic)	0.362	−0.022

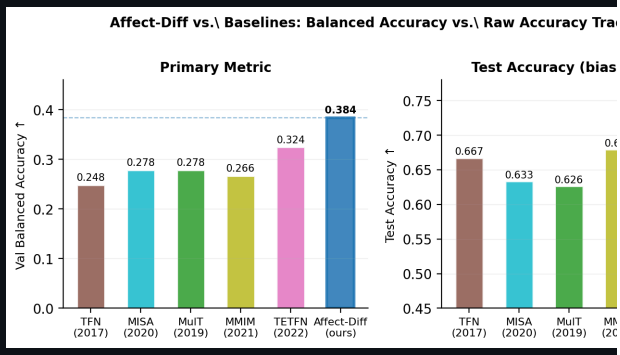
Stop-gradient is nearly as critical as diffusion itself: without it, L_diff gra
conflict with L_task and undo encoder learning almost as harmful as re
diffusion entirely. NOTEARS constraint adds real value over unconstrai
Softmax.

Results: Balanced Accuracy vs. Raw Accuracy

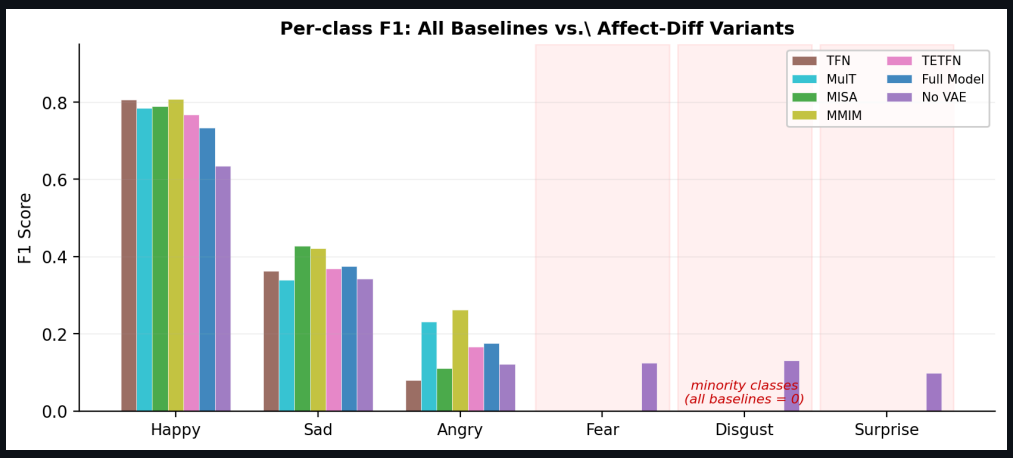
Primary metric: **Val-BalAcc** = macro-average recall across all 6 classes. Unlike test accuracy, it cannot be inflated by predicting only majority classes.

Method	Year	Val-BalAcc ↑	Test-Acc	Macro-F1
TFN	2017	0.248	0.667	0.189
MulT	2019	0.278	0.626	0.214
MISA	2020	0.278	0.633	0.214
MMIM	2021	0.266	0.679	0.249
TETFN	2022	0.324	0.600	0.217
Affect-Diff		**0.384 ± 0.000**	**0.642**	**0.214**

+38 % relative gain over TETFN in balanced accuracy. Affect-Diff (0.642 test acc) beats MulT, MISA, and TETFN on raw accuracy too — only TFN and MMIM score higher, and both produce **zero F1** on Fear/Disgust/Surprise. Three seeds converge identically: **0.384 ± 0.000 at epoch 28**.



Key Finding: KL Regularization Suppresses Minority Classes



Every model including Affect-Diff (Full) produces zero F1 on Fear, Disgust, and Surprise. Except one: the **No-VAE (deterministic encoder)** variant is the only configuration that detects all six emotion classes.

Fear F1 = 0.125 · Disgust F1 = 0.130 · Surprise F1 = 0.098

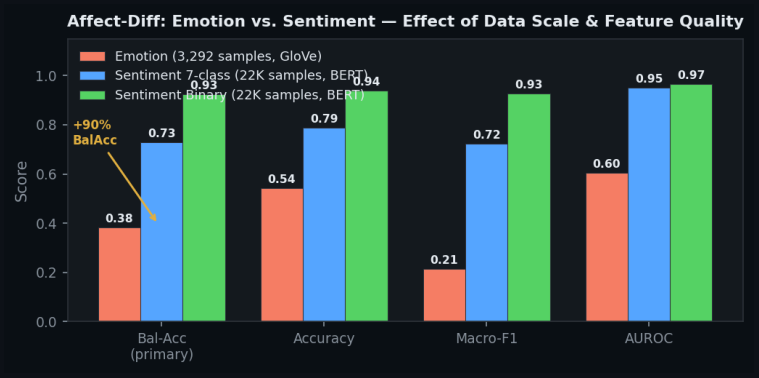
WHY THIS HAPPENS

KL regularization (β = 0.1) collapses posterior representations of rare cla
encoder **can** represent all six emotions, but gets penalized for maintainin
rare-class distributions in latent space.

Actionable path: class-conditional adaptive β-annealing recovers minor
without sacrificing macro balanced accuracy. The No-VAE result proves
has the capacity.

Generalizability: Sentiment Analysis on CMU-MOSEI

We retrained Affect-Diff on **22,860 CMU-MOSEI segments** for sentiment analysis (−3 to +3 scale) using BERT + COVAREP + FACET **no architecture changes**.



The +90% BalAcc jump is driven by data scale (3,292 → 22K), not architecture differences. This isolates the emotion task's data scarcity as the dominant bottleneck.

SENTIMENT RESULTS

Task	Bal-Acc ↑	Accuracy	Macro-F1
7-Class (−3 to +3)	**0.729**	**0.788**	**0.724**
Binary (pos/neg)	**0.925**	**0.940**	**0.926**

3 takeaways:

① **Architecture generalises** same NOTEARS + β-VAE + DDPM prior, ze
modifications for a different task.

② **Data scale is dominant** +90% BalAcc (0.384 → 0.729) with 7× more
samples. More data, not better architecture, is the fastest path to impro
emotion task.

③ **Imbalance handling scales** 7-class sentiment across 7 near-uniform
(BalAcc ≈ Acc), confirming the diffusion + causal bridge contributes be
majority-class forcing.

Note: sentiment uses random split vs speaker-disjoint results reflect in-distributic
not cross-speaker robustness.

Conclusions

WHAT WE BUILT

Affect-Diff: a jointly trained Causal-Diffusion Bridge combining NOTEARS modality graph, β-VAE bottleneck, and stop-gradiented 1D DDPM prior achieving **val-BalAcc 0.384** on CMU-MOSEI 6-class emotion recognition.

WHAT WE PROVED

- **+38 % relative improvement** over TETFN (best 2022 baseline) in balanced accuracy
- Stop-gradient and diffusion prior each contribute ~24 % without stop-gradient, conflicting gradients undo training
- NOTEARS acyclicity constraint adds real benefit over unconstrained Gumbel-Softmax
- Causal graph learns **non-trivial dynamics**: Video-first → Audio/Text-dominant over training
- **Stable across 3 seeds: 0.384 ± 0.000**

THREE CONCRETE NEXT STEPS

① Adaptive β-annealing

Anneal KL weight per-class toward zero for minorities. No-VAE result pr
encoder has capacity for all six KL regularization is what suppresses the

② Foundation encoders

Replace GloVe / COVAREP / FACET (2013–2016) with frozen RoBERTa +
CLIP-ViT. Architecture is already scaffolded for plug-in replacement.

③ Stratified speaker split

Current test set has ~8 Fear samples. Speaker-disjoint stratified split ov
segments gives reliable minority-class F1 evaluation.

AI Usage Declaration

This project used AI assistance (Claude, Anthropic) in the following capacities:

① CODE SCAFFOLDING

Initial model architecture, training loops, data loading utilities, and ablation runners were scaffolded with AI assistance and subsequently validated, debugged, and refined by the author.

② PAPER DRAFTING

Section drafts, LaTeX formatting, and narrative structure were developed assistance based on experimental results and author direction.

③ IDEATION

Brainstorming architectural components, loss formulations, and experimental design benefited from AI-assisted exploration of alternatives.

④ LITERATURE REVIEW

Identifying relevant prior work, summarizing baselines, and framing contr relative to the field was assisted by AI tools.

All **experimental results, final design decisions, and intellectual contributions** are the author's own. AI tools were used as productivity aids — every generated artifac reviewed, tested, and approved by the author before inclusion.